Def. Let X_i $(i \in I)$ be Topological spaces with Topology $\mathcal{T}_i$, respectively. Define a Product subbasis on $\prod_{i \in I} X_i$ as:

$$S_p := \{ p_i^{-1}[\mathcal{U}_i] \mid i \in I,\ \mathcal{U}_i \in \mathcal{T}_i \}$$

The Topology $\mathcal{T}_p$ generated by S_p is called a Product Topology on $\prod_{i \in I} X_i$

Clearly, the product Topology is the smallest Topology s.t all Projections are Continuous. Moreover, on the Product Space, all Projections are Continuous Open onto map.

Thm. Let X, Y_i $(i \in I)$ be Topological Spaces. Define a map

$$f : X \to \prod_{i \in I} Y_i \ ; \ x \mapsto (f_i(x))_{i \in I} \quad \text{where } f_i : X \to Y_i$$

Then, f is Continuous if and only if for all $i \in I$, f_i is Continuous.

Proof If f is continuous, then $f_i = p_i \circ f$ is Continuous, being Composition of Continuous. Conversely, let $B \in \beta_p$ be an base element of $\mathcal{T}_p$. Then, by definition,

$$B = \bigcap_{i=1}^{n} p_i^{-1}[\mathcal{U}_i] \quad \text{for some } \mathcal{U}_i \in \mathcal{T}_i,\ i = 1, \cdots, n.$$

Now,

$$f^{-1}[B] = f^{-1}\left[\bigcap_{i=1}^{n} p_i^{-1}[\mathcal{U}_i]\right] = \bigcap_{i=1}^{n} (f^{-1} \circ p_i^{-1})[\mathcal{U}_i] = \bigcap_{i=1}^{n} (p_i \circ f)^{-1}[\mathcal{U}_i] = \bigcap_{i=1}^{n} f_i[\mathcal{U}_i].$$

$\blacksquare$

「Thm. Let X_i $(i\in I)$ be Topological spaces, $A_i \subset X_i$. Then,

1) $\overline{\prod_{i\in I} A_i} = \prod_{i\in I} \overline{A_i}$

2) $\left(\prod_{i\in I} A_i\right)^\circ \subseteq \prod_{i\in I} A_i^\circ$, the equality holds when the product is finite.

Proof. $(x_i) \in \overline{\prod_{i\in I} A_i}$ $\Leftrightarrow$ $\forall \mathcal{U} \in \beta_p$ with $(x_i)\in\mathcal{U}$, $\mathcal{U} \cap \prod_{i\in I} A_i \neq \phi$

$\Leftrightarrow$ $\forall \mathcal{U} \in \beta_p$ with $(x_i)\in\mathcal{U}$, $\prod_{i\in I} \pi_i[\mathcal{U}] \cap \prod_{i\in I} A_i = \prod_{i\in I} \pi_i[\mathcal{U}] \cap A_i \neq \phi$

$\Leftrightarrow$ $\forall i\in I$, $\forall \mathcal{U}_i \in \beta_i$ with $x_i \in \mathcal{U}_i$, $\mathcal{U}_i \cap A_i \neq \phi$

$\Leftrightarrow$ $(x_i) \in \prod_{i\in I} \overline{A_i}$

(I finite)

$(x_i) \in \left(\prod_{i\in I} A_i\right)^\circ \overset{(*)}{\Leftrightarrow} \exists \mathcal{U} \in \beta_p$ s.t. $(x_i) \in \mathcal{U} \subseteq \prod_{i\in I} A_i$

$\Leftrightarrow$ $\forall i\in I$, $x_i \in P_i[\mathcal{U}] \subseteq A_i$

$\Leftrightarrow$ $\forall i\in I$, $x_i \in A_i^\circ$.

But, if I is Not finite, $(*)$ is broke.